Namespace ExBrainSdk

Classes

ExBrainApi

ExBrain low-level API クラス / ExBrain low-level API class

Delegates

ExBrainApi.NotifyMeasureDatasCallBack

装置計測データ通知用コールバック関数 / Callback for device measurement data notification

ExBrainApi.NotifyProcessMessageCallBack

テキストメッセージ通知用コールバック関数 / Callback for text message notification

ExBrainApi.NotifySearchDevicesCallBack

装置検索結果通知用コールバック関数 / Callback for device search result notification

ExBrainApi.NotifyStatusChangedCallBack

状態変化通知用コールバック関数 / Callback for status change notification

Class ExBrainApi

Namespace: ExBrainSdk

Assembly: ExBrainSdk.dll

ExBrain low-level API クラス / ExBrain low-level API class

```
public class ExBrainApi
```

Inheritance

object ← ExBrainApi

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ExBrainApi(NotifyStatusChangedCallBack?, NotifyProcessMessageCallBack?, bool)

ExBrainApi インスタンスを生成する / Creates an ExBrainApi instance

```
public ExBrainApi(ExBrainApi.NotifyStatusChangedCallBack? notifyStatusChangedCallBack =
null, ExBrainApi.NotifyProcessMessageCallBack? notifyProcessMessageCallBack = null, bool
waitReconnectionWhenLost = false)
```

Parameters

`notifyStatusChangedCallBack` ExBrainApi.NotifyStatusChangedCallBack

状態変化通知用コールバック / Callback for status change notification

`notifyProcessMessageCallBack` ExBrainApi.NotifyProcessMessageCallBack

テキストメッセージ通知用コールバック / Callback for text message notification

`waitReconnectionWhenLost` bool

切断時再接続フラグ / Reconnect on disconnect flag

Properties

Calibration

現状のキャリブレーション情報の参照 / Current calibration information reference

```
public ExBrainCalibration? Calibration { get; }
```

Property Value

ExBrainCalibration

ConnectType

接続方式 / Connection type

```
public EnmSdkConnectType ConnectType { get; }
```

Property Value

EnmSdkConnectType

CurrentStatus

SDK動作状態 / SDK operation status

```
public EnmSdkStatus CurrentStatus { get; }
```

Property Value

EnmSdkStatus

DiscoveredDevice

装置検索結果リスト / Device search result list

```
public ExBrainDiscoveredDevice DiscoveredDevice { get; }
```

Property Value

ExBrainDiscoveredDevice

Gain

現在ゲイン値参照 / Current gain value reference

```
public ExBrainGainInfo? Gain { get; }
```

Property Value

ExBrainGainInfo

HandleId

装置接続ID / Device connection ID

```
public string HandleId { get; }
```

Property Value

string

HwInfo

ハードウェア情報データ / Hardware information data

```
public DeviceHwInfo? HwInfo { get; }
```

Property Value

DeviceHwInfo

HwState

ハードウェア状態データ / Hardware state data

```
public DeviceHwState? HwState { get; }
```

Property Value

DeviceHwState

IsConnected

装置接続済み状態 / Device connected state

```
public bool IsConnected { get; }
```

Property Value

bool

IsInitiated

SDK初期化済み状態 / SDK initialized state

```
public bool IsInitiated { get; }
```

Property Value

bool

LogFolder

ログ出力先フォルダ / Log output folder

```
public string? LogFolder { get; }
```

Property Value

string

ProductName

SDK製品名称 / SDK product name

```
public static string ProductName { get; }
```

Property Value

string

ProductVersion

SDK製品バージョン / SDK product version

```
public static string ProductVersion { get; }
```

Property Value

string

SettingsFolder

設定フォルダ / Settings folder

```
public string? SettingsFolder { get; }
```

Property Value

string

Tag

タグ / Tag

```
public string? Tag { get; set; }
```

Property Value

string

Methods

ConnectDeviceAsync(string, EnmSdkConnectType)

装置接続 / Connect device

```
public Task<EnmSdkResult> ConnectDeviceAsync(string handleId, EnmSdkConnectType connectType)
```

Parameters

`handleId` string

装置接続ID / Device ID for connection

`connectType` EnmSdkConnectType

接続方式 / Connection type

Returns

Task<EnmSdkResult>

結果ｺｰﾄﾞ / Result code

DisconnectDeviceAsync()

装置切断 / Disconnect device

```
public Task DisconnectDeviceAsync()
```

Returns

Task

ExitApi()

API終了 / API termination

```
public void ExitApi()
```

GetBatteryStateAsync(string)

バッテリー状態取得 / Get battery state

```
public Task<BatteryState?> GetBatteryStateAsync(string handleId)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

Returns

Task<BatteryState>

バッテリー状態データ / Battery state data

InitApi(bool, bool, string?, string?, string?)

API初期化 / API initialization

```
[SuppressMessage("CodeQuality", "IDE0060:Remove unused parameter", Justification =
"needOutputCommLog, settingsDir, and tag is not implemented yet")]
```

```
public EnmSdkResult InitApi(bool needOutputTraceLog = false, bool needOutputCommLog = false,
string? logDir = null, string? settingsDir = null, string? tag = null)
```

Parameters

`needOutputTraceLog` bool

トレースログ出力有無 / Output trace log flag

`needOutputCommLog` bool

通信ログ出力有無 / Output communication log flag

`logDir` string

ログフォルダパス / Log folder path

`settingsDir` string

設定フォルダパス / Settings folder path

`tag` string

タグ / Tag

Returns

EnmSdkResult

結果コード / Result code

Remarks

[1.1.5] modify 引数にtag追加 / [1.1.5] added 'tag' argument

PerformCalibrationAsync(string)

キャリブレーション開始 / Start calibration

```
public Task<EnmSdkResult> PerformCalibrationAsync(string handleId)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

Returns

Task<EnmSdkResult>

結果コード / Result code

PerformGainAdjustmentAsync(string, string)

ゲイン調整開始 / Start gain adjustment

```
public Task<EnmSdkResult> PerformGainAdjustmentAsync(string handleId, string
mgcParamLogFolder = "")
```

Parameters

`handleId` string

装置接続ID / Device connection ID

`mgcParamLogFolder` string

ゲイン調整ログフォルダ / Gain adjustment log folder

Returns

Task<EnmSdkResult>

結果コード / Result code

RequestHardwareInfoAsync(string, EnmSdkDeviceInfo)

装置からハードウェア情報を取得 / Get hardware info from device

```
public Task<EnmSdkResult> RequestHardwareInfoAsync(string handleId, EnmSdkDeviceInfo type)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

`type` EnmSdkDeviceInfo

デバイス情報区分 / Device info type

Returns

Task<EnmSdkResult>

結果コード / Result code

RequestHardwareStateAsync(string, EnmSdkDeviceState)

装置からハードウェア状態を取得 / Get hardware state from device

```
public Task<EnmSdkResult> RequestHardwareStateAsync(string handleId, EnmSdkDeviceState type)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

`type` EnmSdkDeviceState

デバイス状態区分 / Device state type

Returns

Task<EnmSdkResult>

結果コード / Result code

ResetBluetoothModuleAsync()

PC内蔵もしくは外付けのBluetooth機能をリセットする / Reset built-in or external Bluetooth function

```
public Task ResetBluetoothModuleAsync()
```

Returns

Task

ResetToDefaultParametersAsync(string)

デフォルト設定にリセットする / Reset to default parameters

```
public Task ResetToDefaultParametersAsync(string handleId)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

Returns

Task

ResetToDefaultParametersFullAsync(string, EnmSdkDeviceType, string)

全機能をデフォルト設定へリセットする / Reset all functions to default

```
public Task ResetToDefaultParametersFullAsync(string handleId, EnmSdkDeviceType deviceType,
string lotId = "")
```

Parameters

`handleId` string

装置接続ID / Device connection ID

`deviceType` EnmSdkDeviceType

デバイス機種 / Device type

`lotId` string

ロットID / Lot ID

Returns

Task

SearchAndConnectDeviceAsync(EnmSdkConnectType, int)

装置の検索と接続 / Device search and connect

```
public Task<EnmSdkResult> SearchAndConnectDeviceAsync(EnmSdkConnectType connectType,
int timeoutSecond)
```

Parameters

`connectType` EnmSdkConnectType

接続方式 / Connection type

`timeoutSecond` int

検索タイムアウト（秒） / Search timeout (seconds)

Returns

Task <EnmSdkResult>

接続結果 / Connection result

SetBlockMarker(string, bool)

ブロックマーカーを設定する / Set block marker

```
public void SetBlockMarker(string markerText, bool markerActive)
```

Parameters

`markerText` string

マーカー文字列 / Marker string

`markerActive` bool

マーカー有効/無効 / Marker enabled/disabled

SetBluetoothEnableAsync(bool)

PC内蔵もしくは外付けのBluetooth機能をON/OFFする / Enable/disable built-in or external Bluetooth function

```
public Task SetBluetoothEnableAsync(bool onOff)
```

Parameters

`onOff` bool

T:ON, F:OFF / T:ON, F:OFF

Returns

Task

SetDeviceNameAsync(string, string)

デバイス機器名称更新 / Update device name

```
public Task<EnmSdkResult> SetDeviceNameAsync(string handleId, string devname)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

`devname` string

デバイス名称(最大8文字) / Device name (max 8 chars)

Returns

Task<EnmSdkResult>

結果コード / Result code

SetEventMarker(string)

イベントマーカーを設定する / Set event marker

```
public void SetEventMarker(string markerText)
```

Parameters

`markerText` string

マーカー文字列 / Marker string

SetIRInitSettingAsync(string, int[], int[], int[], int[])

待機時IR設定の更新 / Update IR settings during standby

```
public Task<EnmSdkResult> SetIRInitSettingAsync(string handleId, int[] swGains, int[]
pwGains, int[] ioValues, int[] brights)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

`swGains` int[]

SWゲイン配列 / SW gain array

`pwGains` int[]

PWゲイン配列 / PW gain array

`ioValues` int[]

IO値配列 / IO values array

`brights` int[]

明るさ配列 / Brightness array

Returns

Task<EnmSdkResult>

結果コード / Result code

SetMgcGainAsync(string, int, int, int, int)

ゲイン値更新 / Update gain values

```
public Task<EnmSdkResult> SetMgcGainAsync(string handleId, int gainL1, int gainL3, int
gainR1, int gainR3)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

`gainL1` int

左L1ゲイン / Left L1 gain

`gainL3` int

左L3ゲイン / Left L3 gain

`gainR1` int

右R1ゲイン / Right R1 gain

`gainR3` int

右R3ゲイン / Right R3 gain

Returns

Task<EnmSdkResult>

結果コード / Result code

SetMgcLedAsync(string, int, int)

赤外線光量値更新 / Update infrared LED brightness

```
public Task<EnmSdkResult> SetMgcLedAsync(string handleId, int ledBrightnessL,
int ledBrightnessR)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

`ledBrightnessL` int

左LED光量 / Left LED brightness

`ledBrightnessR` int

右LED光量 / Right LED brightness

Returns

Task<EnmSdkResult>

結果コード / Result code

SetMotionRangeAccAsync(string, EnmSdkAccRange)

モーション加速度レンジ更新 / Update motion acceleration range

```
public Task<EnmSdkResult> SetMotionRangeAccAsync(string handleId, EnmSdkAccRange range)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

`range` EnmSdkAccRange

モーション加速度レンジ区分 / Motion acceleration range type

Returns

Task<EnmSdkResult>

結果コード / Result code

SetMotionRangeGyroAsync(string, EnmSdkGyroRange)

モーション角速度レンジ更新 / Update motion gyro range

```
public Task<EnmSdkResult> SetMotionRangeGyroAsync(string handleId, EnmSdkGyroRange range)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

`range` EnmSdkGyroRange

モーション角速度レンジ区分 / Motion gyro range type

Returns

Task<EnmSdkResult>

結果コード / Result code

SetUsbRTOutputModeAsync(string, bool)

リアルタイムUSB出力機能更新 / Update real-time USB output mode

```
public Task<EnmSdkResult> SetUsbRTOutputModeAsync(string handleId, bool output)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

`output` bool

T=使用／F=未使用 / T=enabled/F=disabled

Returns

Task<EnmSdkResult>

結果コード / Result code

StartMeasureAsync(string, string?, NotifyMeasureDatasCallBack, bool, EnmSdkMeasurementCycle, EnmSdkMeasurementMode)

自動計測開始 / Start automatic measurement

```
public Task<EnmSdkResult> StartMeasureAsync(string handleId, string? measureLogFolder,
ExBrainApi.NotifyMeasureDatasCallBack callback, bool recalibrate = false,
EnmSdkMeasurementCycle measurementCycle = EnmSdkMeasurementCycle.e10Hz,
EnmSdkMeasurementMode measurementMode = EnmSdkMeasurementMode.eNormal)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

`measureLogFolder` string

計測結果出力先フォルダパス / Measurement log folder path

`callback` ExBrainApi.NotifyMeasureDatasCallBack

計測データ通知用コールバック / Callback for measurement data notification

`recalibrate` bool

再キャリブレーションフラグ / Recalibration flag

`measurementCycle` EnmSdkMeasurementCycle

計測サイクル / Measurement cycle

`measurementMode` EnmSdkMeasurementMode

計測モード / Measurement mode

Returns

Task <EnmSdkResult>

結果コード / Result code

StartSearchDevice(ExBrainSearchParameter, NotifySearchDevicesCallBack?)

装置検索開始 / Start device search

```
public void StartSearchDevice(ExBrainSearchParameter param,
ExBrainApi.NotifySearchDevicesCallBack? callback = null)
```

Parameters

`param` ExBrainSearchParameter

検索パラメータ / Search parameter

`callback` ExBrainApi.NotifySearchDevicesCallBack

検索結果通知用コールバック / Callback for search result notification

StopMeasureAsync(string)

自動計測停止 / Stop measurement

```
public Task<EnmSdkResult> StopMeasureAsync(string handleId)
```

Parameters

`handleId` string

装置接続ID / Device connection ID

Returns

Task<EnmSdkResult>

結果ｺｰﾄﾞ / Result code

StopSearchDevice()

装置検索停止 / Stop device search

```
public void StopSearchDevice()
```

Delegate ExBrainApi.NotifyMeasureDatasCallBack

Namespace: ExBrainSdk

Assembly: ExBrainSdk.dll

装置計測データ通知用コールバック関数 / Callback for device measurement data notification

```
public delegate void ExBrainApi.NotifyMeasureDatasCallBack(string id,
ExBrainMeasureData measure_data)
```

Parameters

`id` string

装置接続ID / Device connection ID

`measure_data` ExBrainMeasureData

計測データ / Measurement data

Delegate ExBrainApi.NotifyProcessMessageCallBack

Namespace: ExBrainSdk

Assembly: ExBrainSdk.dll

テキストメッセージ通知用コールバック関数 / Callback for text message notification

```
public delegate void ExBrainApi.NotifyProcessMessageCallBack(string id, string message,
EnmSdkMessageType type)
```

Parameters

`id` string

装置接続ID / Device connection ID

`message` string

メッセージ内容 / Message content

`type` EnmSdkMessageType

メッセージ種別 / Message type

Delegate ExBrainApi.NotifySearchDevicesCallBack

Namespace: ExBrainSdk

Assembly: ExBrainSdk.dll

装置検索結果通知用コールバック関数 / Callback for device search result notification

```
public delegate void ExBrainApi.NotifySearchDevicesCallBack(EnmSdkSearchEvent ev,
DeviceInformationEx device)
```

Parameters

`ev` EnmSdkSearchEvent

検索イベント / Search event

`device` DeviceInformationEx

デバイス情報 / Device information

Delegate ExBrainApi.NotifyStatusChangedCallBack

Namespace: ExBrainSdk

Assembly: ExBrainSdk.dll

状態変化通知用コールバック関数 / Callback for status change notification

```
public delegate void ExBrainApi.NotifyStatusChangedCallBack(string id, EnmSdkStatus status)
```

Parameters

`id` string

装置接続ID / Device connection ID

`status` EnmSdkStatus

状態値 / Status value

Namespace ExBrainSdk.Datas

Classes

BatteryState

バッテリー状態 / Battery state

DeviceHwInfo

ハードウェア装置情報 / Hardware device information

DeviceHwState

ハードウェア装置状態 / Device hardware state

DeviceInformationEx

拡張装置情報データ / Extended device information

ExBrainCalibration

脳計測キャリブレーションデータ / Brain measurement calibration data

ExBrainDiscoveredDevice

発見された装置データ / Discovered device information

ExBrainGainInfo

ゲイン情報 / Gain information

ExBrainMeasureData

装置計測結果データ / Device measurement result data

ExBrainSearchParameter

装置検索パラメータ / Device search parameters

Class BatteryState

Namespace: ExBrainSdk.Datas

Assembly: ExBrainSdk.dll

バッテリー状態 / Battery state

```
public class BatteryState
```

Inheritance

object ← BatteryState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

BatteryGauge

電池残量 [%] / Battery gauge [%]

```
public double BatteryGauge { get; }
```

Property Value

double

BatteryVolt

電池電圧 [mV] / Battery voltage [mV]

```
public double BatteryVolt { get; }
```

Property Value

double

Class DeviceHwInfo

Namespace: ExBrainSdk.Datas

Assembly: ExBrainSdk.dll

ハードウェア装置情報 / Hardware device information

```
public class DeviceHwInfo
```

Inheritance

object ← DeviceHwInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

DeviceFwVersion

ファームウェアバージョン / Firmware version

```
public string DeviceFwVersion { get; }
```

Property Value

string

DeviceId

機器 ID / Device ID

```
public string DeviceId { get; }
```

Property Value

string

DeviceName

機器名称 / Device name

```
public string DeviceName { get; }
```

Property Value

string

DeviceType

機種区分 / Device type

```
public EnmSdkDeviceType DeviceType { get; }
```

Property Value

EnmSdkDeviceType

EnabledOutputUsb

USB リアルタイム出力機能有無 / USB real-time output enabled flag

```
public bool EnabledOutputUsb { get; }
```

Property Value

bool

LotId

ロット ID / Lot ID

```
public string LotId { get; }
```

Property Value

string

RangeAcc

モーション加速度レンジ / Motion acceleration range

```
public EnmSdkAccRange RangeAcc { get; }
```

Property Value

EnmSdkAccRange

RangeGyro

モーション角速度レンジ / Motion gyro range

```
public EnmSdkGyroRange RangeGyro { get; }
```

Property Value

EnmSdkGyroRange

ResoluteAcc

モーション加速度分解能 ×10^-3 [G/bit] / Motion acceleration resolution ×10^-3 [G/bit]

```
public double ResoluteAcc { get; }
```

Property Value

double

ResoluteGyro

モーション角速度分解能 [DPS/bit] / Motion gyro resolution [DPS/bit]

```
public double ResoluteGyro { get; }
```

Property Value

double

ResolutePd

PD 計測値分解能 [mV/bit] / PD resolution [mV/bit]

```
public double ResolutePd { get; }
```

Property Value

double

SamplingBattery

バッテリーサンプリング / Battery sampling rate

```
public EnmSdkSamplingBattery SamplingBattery { get; }
```

Property Value

EnmSdkSamplingBattery

SamplingMotion

モーションサンプリング / Motion sampling rate

```
public EnmSdkSamplingMotion SamplingMotion { get; }
```

Property Value

EnmSdkSamplingMotion

TransRateBattery

バッテリーデータ転送周期 / Battery transfer rate

```
public EnmSdkTransRateBattery TransRateBattery { get; }
```

Property Value

EnmSdkTransRateBattery

TransRateHR

心拍数データ転送周期 / Heart-rate transfer rate

```
public EnmSdkTransRateHR TransRateHR { get; }
```

Property Value

EnmSdkTransRateHR

TransRateMotion

モーションデータ転送周期 / Motion transfer rate

```
public EnmSdkTransRateMotion TransRateMotion { get; }
```

Property Value

EnmSdkTransRateMotion

TransRatePD

脳活動計測データ転送周期 / Brain activity transfer rate

```
public EnmSdkTransRatePD TransRatePD { get; }
```

Property Value

EnmSdkTransRatePD

VMaxPd

PD 計測値デジット最大値 / Maximum PD digit value

```
public int VMaxPd { get; }
```

Property Value

int

Class DeviceHwState

Namespace: ExBrainSdk.Datas

Assembly: ExBrainSdk.dll

ハードウェア装置状態 / Device hardware state

```
public class DeviceHwState
```

Inheritance

object ← DeviceHwState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

BatteryTemperature

バッテリー温度 [℃] / Battery temperature [°C]

```
public double BatteryTemperature { get; }
```

Property Value

double

CpuTemperature

CPU 温度 [℃] / CPU temperature [°C]

```
public double CpuTemperature { get; }
```

Property Value

double

EnabledBtComm

Bluetooth 通信接続状態 / Bluetooth communication connection state

```
public bool EnabledBtComm { get; }
```

Property Value

bool

EnabledBtModule

Bluetooth モジュール使用状態 / Bluetooth module enabled state

```
public bool EnabledBtModule { get; }
```

Property Value

bool

EnabledGauge

残量ゲージセンサ検出状態 / Battery gauge sensor detection state

```
public bool EnabledGauge { get; }
```

Property Value

bool

EnabledMotion

モーションセンサ検出状態 / Motion sensor detection state

```
public bool EnabledMotion { get; }
```

Property Value

bool

EnabledPdL1

PD-L1 値検出状態 / PD-L1 detection state

```
public bool EnabledPdL1 { get; }
```

Property Value

bool

EnabledPdL3

PD-L3 値検出状態 / PD-L3 detection state

```
public bool EnabledPdL3 { get; }
```

Property Value

bool

EnabledPdR1

PD-R1 値検出状態 / PD-R1 detection state

```
public bool EnabledPdR1 { get; }
```

Property Value

bool

EnabledPdR3

PD-R3 値検出状態 / PD-R3 detection state

```
public bool EnabledPdR3 { get; }
```

Property Value

bool

EnabledSensModule

センサモジュール検出状態 / Sensor module detection state

```
public bool EnabledSensModule { get; }
```

Property Value

bool

EnabledUsbComm

USB 通信接続状態 / USB communication connection state

```
public bool EnabledUsbComm { get; }
```

Property Value

bool

EnabledUsbVbus

USB VBUS 状態 / USB VBUS state

```
public bool EnabledUsbVbus { get; }
```

Property Value

bool

LedBrightnessL

IR 光量値（L） / IR brightness value (L)

```
public int LedBrightnessL { get; }
```

Property Value

int

LedBrightnessR

IR 光量値（R） / IR brightness value (R)

```
public int LedBrightnessR { get; }
```

Property Value

int

LedEnableL

IR IO 出力状態（L） / IR IO output state (L)

```
public bool LedEnableL { get; }
```

Property Value

bool

LedEnableR

IR IO 出力状態（R） / IR IO output state (R)

```
public bool LedEnableR { get; }
```

Property Value

bool

MotionTemperature

モーション温度 [℃] / Motion temperature [°C]

```
public double MotionTemperature { get; }
```

Property Value

double

Class DeviceInformationEx

Namespace: ExBrainSdk.Datas

Assembly: ExBrainSdk.dll

拡張装置情報データ / Extended device information

```
public class DeviceInformationEx
```

Inheritance

object ← DeviceInformationEx

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ConnectType

接続方式 / Connection type

```
public EnmSdkConnectType ConnectType { get; }
```

Property Value

EnmSdkConnectType

DeviceId

接続 ID / Device connection ID

```
public string DeviceId { get; }
```

Property Value

string

IsPaired

Bluetooth ペアリング済みフラグ / Bluetooth paired flag

```
public bool IsPaired { get; }
```

Property Value

bool

TargetName

接続対象の名称 / Target device name

```
public string TargetName { get; }
```

Property Value

string

TargetType

接続対象の機種 / Target device type

```
public EnmSdkDeviceType TargetType { get; }
```

Property Value

EnmSdkDeviceType

UsbPort

USB の COM ポート情報 / USB COM port information

```
public string UsbPort { get; }
```

Property Value

string

UsbVidPid

USB の ID 情報 / USB VID/PID information

```
public string UsbVidPid { get; }
```

Property Value

string

Class ExBrainCalibration

Namespace: ExBrainSdk.Datas

Assembly: ExBrainSdk.dll

脳計測キャリブレーションデータ / Brain measurement calibration data

```
public class ExBrainCalibration
```

Inheritance

object ← ExBrainCalibration

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

HbDensityOffsetL1

左 1cm のヘモグロビン密度オフセット / Left 1cm hemoglobin density offset

```
public double HbDensityOffsetL1 { get; set; }
```

Property Value

double

HbDensityOffsetL3

左 3cm のヘモグロビン密度オフセット / Left 3cm hemoglobin density offset

```
public double HbDensityOffsetL3 { get; set; }
```

Property Value

double

HbDensityOffsetR1

右 1cm のヘモグロビン密度オフセット / Right 1cm hemoglobin density offset

```
public double HbDensityOffsetR1 { get; set; }
```

Property Value

double

HbDensityOffsetR3

右 3cm のヘモグロビン密度オフセット / Right 3cm hemoglobin density offset

```
public double HbDensityOffsetR3 { get; set; }
```

Property Value

double

Class ExBrainDiscoveredDevice

Namespace: ExBrainSdk.Datas

Assembly: ExBrainSdk.dll

発見された装置データ / Discovered device information

```
public class ExBrainDiscoveredDevice
```

Inheritance

object ← ExBrainDiscoveredDevice

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

DeviceCount

発見された装置数 / Number of discovered devices

```
public int DeviceCount { get; }
```

Property Value

int

Devices

発見された装置リスト / Discovered device list

```
public IReadOnlyList<DeviceInformationEx> Devices { get; }
```

Property Value

IReadOnlyList <DeviceInformationEx>

Class ExBrainGainInfo

Namespace: ExBrainSdk.Datas

Assembly: ExBrainSdk.dll

ゲイン情報 / Gain information

```
public class ExBrainGainInfo
```

Inheritance

object ← ExBrainGainInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ExBrainGainInfo()

コンストラクタ / Constructor

```
public ExBrainGainInfo()
```

Properties

Gains

ゲイン設定値 / Gain values

```
public int[] Gains { get; set; }
```

Property Value

int[]

Remarks

0:L1cm 1-16 1:L3cm 1-64 2:R1cm 1-16 3:R3cm 1-64 / 0:L1cm 1-16 1:L3cm 1-64 2:R1cm 1-16 3:R3cm 1-64

Methods

Initialize(EnmSdkDeviceType)

初期化 / Initialize gain values

```
public void Initialize(EnmSdkDeviceType devType)
```

Parameters

`devType` EnmSdkDeviceType

デバイス機種 / Device type

Class ExBrainMeasureData

Namespace: ExBrainSdk.Datas

Assembly: ExBrainSdk.dll

装置計測結果データ / Device measurement result data

```
public class ExBrainMeasureData
```

Inheritance

object ← ExBrainMeasureData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

AccelerationX

X 軸加速度 [G] / X-axis acceleration [G]

```
[JsonInclude]
public double AccelerationX { get; }
```

Property Value

double

AccelerationY

Y 軸加速度 [G] / Y-axis acceleration [G]

```
[JsonInclude]
public double AccelerationY { get; }
```

Property Value

double

AccelerationZ

Z 軸加速度 [G] / Z-axis acceleration [G]

```
[JsonInclude]
public double AccelerationZ { get; }
```

Property Value

double

AngularVelocityX

X 軸角速度 [DPS] / X-axis angular velocity [DPS]

```
[JsonInclude]
public double AngularVelocityX { get; }
```

Property Value

double

AngularVelocityY

Y 軸角速度 [DPS] / Y-axis angular velocity [DPS]

```
[JsonInclude]
public double AngularVelocityY { get; }
```

Property Value

double

AngularVelocityZ

Z 軸角速度 [DPS] / Z-axis angular velocity [DPS]

```
[JsonInclude]
public double AngularVelocityZ { get; }
```

Property Value

double

BrainIndexValueL

左側脳活動指標値 / Left brain activity index value

```
[JsonInclude]
public double BrainIndexValueL { get; }
```

Property Value

double

BrainIndexValueR

右側脳活動指標値 / Right brain activity index value

```
[JsonInclude]
public double BrainIndexValueR { get; }
```

Property Value

double

ColorIndexLeft

脳活動指標値（左） / Brain activity index (left)

```
[JsonInclude]
public int ColorIndexLeft { get; }
```

Property Value

int

ColorIndexRight

脳活動指標値（右） / Brain activity index (right)

```
[JsonInclude]
public int ColorIndexRight { get; }
```

Property Value

int

DropCount

前データとの通番抜けカウント / Missing sequence count from previous data

```
[JsonInclude]
public int DropCount { get; }
```

Property Value

int

ElapsedSeconds

計測経過時間 [sec] / Elapsed measurement time [sec]

```
[JsonInclude]
public double ElapsedSeconds { get; }
```

Property Value

double

HbDensityL1

ヘモグロビン密度 L1 / Hemoglobin Density L1

```
[JsonInclude]
public double HbDensityL1 { get; }
```

Property Value

double

HbDensityL3

ヘモグロビン密度 L3 / Hemoglobin Density L3

```
[JsonInclude]
public double HbDensityL3 { get; }
```

Property Value

double

HbDensityR1

ヘモグロビン密度 R1 / Hemoglobin Density R1

```
[JsonInclude]
public double HbDensityR1 { get; }
```

Property Value

double

HbDensityR3

ヘモグロビン密度 R3 / Hemoglobin Density R3

```
[JsonInclude]
public double HbDensityR3 { get; }
```

Property Value

double

HeartRate

推定心拍数 [bpm] / Estimated heart rate [bpm]

```
[JsonInclude]
public double HeartRate { get; }
```

Property Value

double

MarkerText

マーカーテキスト / Marker text

```
[JsonInclude]
public string MarkerText { get; }
```

Property Value

string

SerialCounter

計測通番カウンタ / Measurement sequence counter

```
[JsonInclude]
public double SerialCounter { get; }
```

Property Value

double

SyntheticAcceleration

合成加速度 [G] / Synthetic acceleration [G]

```
[JsonInclude]
public double SyntheticAcceleration { get; }
```

Property Value

double

UpdateDateTime

更新日時（ローカル時刻） / Updated datetime (local time)

```
[JsonInclude]
public DateTime UpdateDateTime { get; }
```

Property Value

DateTime

Class ExBrainSearchParameter

Namespace: ExBrainSdk.Datas

Assembly: ExBrainSdk.dll

装置検索パラメータ / Device search parameters

```
public class ExBrainSearchParameter
```

Inheritance

object ← ExBrainSearchParameter

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Keywords

検索キーワード（複数可） / Search keywords (multiple allowed)

```
public string[] Keywords { get; set; }
```

Property Value

string[]

ShouldNonPairedBluetooth

周辺 Bluetooth を検索対象にする / Include non-paired Bluetooth devices in search

```
public bool ShouldNonPairedBluetooth { get; set; }
```

Property Value

bool

ShouldPairedBluetooth

ペアリング済み Bluetooth を検索対象にする / Include paired Bluetooth devices in search

```
public bool ShouldPairedBluetooth { get; set; }
```

Property Value

bool

ShouldUSB

USB を検索対象にする / Include USB devices in search

```
public bool ShouldUSB { get; set; }
```

Property Value

bool

TimeoutSecond

検索タイムアウト [秒] / Search timeout in seconds

```
public int TimeoutSecond { get; set; }
```

Property Value

int

Namespace ExBrainSdk.Enumerations

Enums

EnmSdkAccRange
モーション加速度レンジ区分 / Motion acceleration range categories

EnmSdkConnectType
接続方式定義 / Connection type definitions

EnmSdkDeviceInfo
デバイス情報区分定義 / Device info categories

EnmSdkDeviceState
デバイス状態区分定義 / Device state categories

EnmSdkDeviceType
機種区分 / Device type categories

EnmSdkDiagRecordType
ダイアグレコードタイプ / Diagnostic record types

EnmSdkEvent
動作イベント定義 / SDK event definitions

EnmSdkGyroRange
モーション角速度レンジ区分 / Motion gyro range categories

EnmSdkMeasurementCycle
計測サイクル定義 / Measurement cycle definitions

EnmSdkMeasurementMode
計測モード定義 / Measurement mode definitions

EnmSdkMessageType
テキストメッセージタイプ / Text message types

EnmSdkPdLocations
計測位置区分 / Measurement locations

EnmSdkResult
処理結果定義 / Result code definitions

EnmSdkSamplingBattery
バッテリーサンプリング区分 / Battery sampling rate categories

EnmSdkSamplingMotion

モーションサンプリング区分 / Motion sampling rate categories

EnmSdkSearchEvent

機器検索結果区分 / Device discovery event categories

EnmSdkStatus

動作状態定義 / SDK status definitions

EnmSdkTransRateBattery

バッテリー転送周期区分 / Battery transfer rate categories

EnmSdkTransRateHR

心拍数転送周期区分 / Heart-rate transfer rate categories

EnmSdkTransRateMotion

モーション転送周期区分 / Motion transfer rate categories

EnmSdkTransRatePD

脳活動計測データ転送周期区分 / Brain activity transfer rate categories

Enum EnmSdkAccRange

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

モーション加速度レンジ区分 / Motion acceleration range categories

```
public enum EnmSdkAccRange
```

Fields

`e16G = 4`

16G / 16G

`e2G = 1`

2G / 2G

`e4G = 2`

4G / 4G

`e8G = 3`

8G / 8G

`eDefault = 0`

デフォルト / Default

Enum EnmSdkConnectType

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

接続方式定義 / Connection type definitions

```
public enum EnmSdkConnectType
```

Fields

`eBluetooth = 1`

Bluetooth / Bluetooth

`eNotSelected = 0`

未選択 / Not selected

`eUsb = 2`

USB / USB

Enum EnmSdkDeviceInfo

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

デバイス情報区分定義 / Device info categories

```
public enum EnmSdkDeviceInfo
```

Fields

`eAccSettings = 6`

加速度設定情報 / Acceleration settings

`eAll = 0`

全ての情報 / All information

`eBatteryCapacity = 14`

バッテリー容量設定 / Battery capacity settings

`eDevId = 2`

デバイス ID / Device ID

`eDevName = 3`

デバイス名称 / Device name

`eDevType = 1`

機種 / Device type

`eFwVersion = 12`

ファームウェアバージョン情報 / Firmware version information

`eGyroSettings = 7`

角速度設定情報 / Gyro settings

`eHwSettings = 5`

ハードウェア情報 / Hardware settings

`eInitIrSettings = 13`

起動時近赤外線 LED 設定 / Startup infrared LED settings

`eLotId = 4`

ロット ID / Lot ID

`eMeasureAnalysisSettings = 10`

計測解析設定 / Measurement analysis settings

`eMeasureModeSettings = 8`

計測モード設定 / Measurement mode settings

`eMeasureTransferSettings = 9`

計測転送モード設定 / Measurement transfer settings

`eMinimum = 15`

最低限のデータ取得 / Minimum data acquisition

`eUsbRTOutput = 11`

USB リアルタイム出力設定 / USB real-time output settings

Enum EnmSdkDeviceState

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

デバイス状態区分定義 / Device state categories

```
public enum EnmSdkDeviceState
```

Fields

`eAll = 0`

全ての状態 / All states

`eBatteryGauge = 1`

バッテリー残量状態 / Battery gauge state

`eBatteryState = 10`

バッテリー検知状態 / Battery detection state

`eBatteryVolt = 2`

バッテリー電圧状態 / Battery voltage state

`eBluetoothState = 6`

Bluetooth 状態 / Bluetooth state

`eIrLedState = 4`

近赤外線 LED 状態 / Infrared LED state

`eMotionState = 9`

モーション検知状態 / Motion detection state

`ePDState = 8`

受光センサ検知状態 / Photodiode detection state

`eSensorModuleState = 7`

センサモジュール状態 / Sensor module state

`eTemperatures = 3`

温度状態 / Temperature state

`eUsbState = 5`

USB 状態 / USB state

Enum EnmSdkDeviceType

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

機種区分 / Device type categories

```
public enum EnmSdkDeviceType
```

Fields

`eHOT2000 = 1`

HOT2000 / HOT2000

`eUnknown = -1`

不明 / Unknown

`eXB01 = 2`

XB01 / XB01

Enum EnmSdkDiagRecordType

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

ダイアグレコードタイプ / Diagnostic record types

```
public enum EnmSdkDiagRecordType
```

Fields

`eDiag = 0`

ダイアグレコード / Diagnostic record

`eTrace = 1`

トレースレコード / Trace record

Enum EnmSdkEvent

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

動作イベント定義 / SDK event definitions

```
public enum EnmSdkEvent
```

Fields

`eBluetoothResetCompleted = 5`

Bluetooth モジュールリセット完了 / Bluetooth module reset completed

`eConnectCompleted = 1`

接続完了 / Connection completed

`eDisconnectCompleted = 3`

切断完了 / Disconnection completed

`eDisconnectDetected = 4`

切断検知 / Disconnection detected

`eReconnectCompleted = 2`

再接続完了 / Reconnection completed

Enum EnmSdkGyroRange

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

モーション角速度レンジ区分 / Motion gyro range categories

```
public enum EnmSdkGyroRange
```

Fields

`e1000DPS = 4`

1000DPS / 1000DPS

`e125DPS = 1`

125DPS / 125DPS

`e2000DPS = 5`

2000DPS / 2000DPS

`e250DPS = 2`

250DPS / 250DPS

`e500DPS = 3`

500DPS / 500DPS

`eDefault = 0`

デフォルト / Default

Enum EnmSdkMeasurementCycle

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

計測サイクル定義 / Measurement cycle definitions

```
public enum EnmSdkMeasurementCycle
```

Fields

`e100Hz = 2`

100Hz

`e10Hz = 0`

10Hz

`e500Hz = 3`

500Hz

`e50Hz = 1`

50Hz

Enum EnmSdkMeasurementMode

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

計測モード定義 / Measurement mode definitions

```
public enum EnmSdkMeasurementMode
```

Fields

`eHighPrecisionLowReliability = 2`

高精度・低信頼 / High precision, low reliability

`eLowPrecisionHighReliability = 1`

低精度・高信頼 / Low precision, high reliability

`eNormal = 0`

通常計測 / Normal measurement

Enum EnmSdkMessageType

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

テキストメッセージタイプ / Text message types

```
public enum EnmSdkMessageType
```

Fields

`eError = 2`

エラー / Error

`eNormal = 0`

通常 / Normal

`eStatus = 1`

状態 / Status

Enum EnmSdkPdLocations

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

計測位置区分 / Measurement locations

```
public enum EnmSdkPdLocations
```

Fields

`L1cm = 0`

L1cm / L1cm

`L3cm = 1`

L3cm / L3cm

`R1cm = 2`

R1cm / R1cm

`R3cm = 3`

R3cm / R3cm

Enum EnmSdkResult

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

処理結果定義 / Result code definitions

```
public enum EnmSdkResult
```

Fields

`eCalibratingDataFullErr = 25`

キャリブレーションデータ満杯エラー / Calibration data full error

`eCalibratingStartErr = 23`

キャリブレーション開始エラー / Calibration start error

`eCalibratingStopErr = 24`

キャリブレーション停止エラー / Calibration stop error

`eFailed = 1`

失敗 / Failed

`eFittingDataFullErr = 16`

装着調整データ満杯エラー / Fitting data full error

`eFittingGainOverErr = 17`

装着調整ゲインオーバーエラー / Fitting gain over error

`eFittingLastCheckErr = 19`

装着調整最終チェックエラー / Fitting final check error

`eFittingLastCheckSaturationOverErr = 20`

装着調整最終チェック飽和オーバーエラー / Fitting final check saturation over error

`eFittingLastCheckValueOverErr = 22`

装着調整最終チェック値オーバーエラー / Fitting final check value over error

`eFittingLastCheckValueUnderErr = 21`

装着調整最終チェック値アンダーエラー / Fitting final check value under error

`eFittingRetryOverErr = 18`

装着調整リトライオーバーエラー / Fitting retry over error

`eFittingStartErr = 14`

装着調整開始エラー / Fitting start error

`eFittingStopErr = 15`

装着調整停止エラー / Fitting stop error

`eLicenseError = 28`

ライセンスエラー / License error

`eMeasureStartErr = 26`

計測開始エラー / Measurement start error

`eMismatchDeviceId = 12`

デバイスID不一致 / Device ID mismatch

`eMismatchStatus = 11`

状態不一致 / Status mismatch

`eNotConnected = 4`

未接続 / Not connected

`eNotFoundDevice = 10`

デバイス未検出 / Device not found

`eNotInitiated = 3`

未初期化 / Not initiated

`eNotSupported = 2`

未サポート / Not supported

`eParamWrong = 9`

パラメータ不正 / Invalid parameter

`ePreSameParam = 13`

前回と同じパラメータのため処理なし / No operation because parameters are unchanged from previous call

`eReadFailed = 6`

読込失敗 / Read failed

`eReadTimeout = 5`

読込タイムアウト / Read timeout

`eSuccess = 0`

成功 / Success

`eSystemError = 27`

システムエラー / System error

`eWriteFailed = 8`

書込失敗 / Write failed

`eWriteTimeout = 7`

書込タイムアウト / Write timeout

Enum EnmSdkSamplingBattery

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

バッテリーサンプリング区分 / Battery sampling rate categories

```
public enum EnmSdkSamplingBattery
```

Fields

`e0_1HZ = 1`

0.1Hz / 0.1Hz

`e0_2HZ = 2`

0.2Hz / 0.2Hz

`e0_5HZ = 3`

0.5Hz / 0.5Hz

`e1HZ = 4`

1Hz / 1Hz

`eDisable = 0`

無効 / Disabled

Enum EnmSdkSamplingMotion

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

モーションサンプリング区分 / Motion sampling rate categories

```
public enum EnmSdkSamplingMotion
```

Fields

`e10HZ = 2`

10Hz / 10Hz

`e1HZ = 1`

1Hz / 1Hz

`e20HZ = 3`

20Hz / 20Hz

`e50HZ = 4`

50Hz / 50Hz

`eDisable = 0`

無効 / Disabled

Enum EnmSdkSearchEvent

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

機器検索結果区分 / Device discovery event categories

```
public enum EnmSdkSearchEvent
```

Fields

`eAdded = 0`

追加 / Added

`eRemoved = 2`

削除 / Removed

`eStopped = 3`

停止 / Stopped

`eUpdated = 1`

更新 / Updated

Enum EnmSdkStatus

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

動作状態定義 / SDK status definitions

```
public enum EnmSdkStatus
```

Fields

`eCalibrating = 8`

装置準備中（キャリブレーション中） / Preparing device (calibrating)

`eConnecting = 4`

装置接続中 / Connecting device

`eDisconnecting = 5`

装置切断中 / Disconnecting device

`eFitting = 7`

装着調整中 / Fitting

`eIdle = 6`

待機中 / Idle

`eInitial = 1`

初期化中 / Initializing

`eLosting = 10`

通信ロスト中 / Communication lost

`eMeasureLosting = 11`

計測時通信ロスト中 / Communication lost while measuring

`eMeasuring = 9`

装置計測中 / Measuring

`eOffline = 2`

オフライン / Offline

`eSearching = 3`

装置検索中 / Searching device

Enum EnmSdkTransRateBattery

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

バッテリー転送周期区分 / Battery transfer rate categories

```
public enum EnmSdkTransRateBattery
```

Fields

`e0_1HZ = 1`

0.1Hz / 0.1Hz

`e0_2HZ = 2`

0.2Hz / 0.2Hz

`e0_5HZ = 3`

0.5Hz / 0.5Hz

`e1HZ = 4`

1Hz / 1Hz

`eDisable = 0`

無効 / Disabled

Enum EnmSdkTransRateHR

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

心拍数転送周期区分 / Heart-rate transfer rate categories

```
public enum EnmSdkTransRateHR
```

Fields

`e10HZ = 4`

10Hz / 10Hz

`e1HZ = 1`

1Hz / 1Hz

`e2HZ = 2`

2Hz / 2Hz

`e5HZ = 3`

5Hz / 5Hz

`eDisable = 0`

無効 / Disabled

Enum EnmSdkTransRateMotion

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

モーション転送周期区分 / Motion transfer rate categories

```
public enum EnmSdkTransRateMotion
```

Fields

`e10HZ = 2`

10Hz / 10Hz

`e1HZ = 1`

1Hz / 1Hz

`e20HZ = 3`

20Hz / 20Hz

`e50HZ = 4`

50Hz / 50Hz

`eDisable = 0`

無効 / Disabled

Enum EnmSdkTransRatePD

Namespace: ExBrainSdk.Enumerations

Assembly: ExBrainSdk.dll

脳活動計測データ転送周期区分 / Brain activity transfer rate categories

```
public enum EnmSdkTransRatePD
```

Fields

`e10HZ = 4`

10Hz / 10Hz

`e1HZ = 1`

1Hz / 1Hz

`e2HZ = 2`

2Hz / 2Hz

`e5HZ = 3`

5Hz / 5Hz

`eDisable = 0`

無効 / Disabled